

Research and implementation of an analytical database testing platform in telecommunication industry

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Abstract—In industry, analytic database is known to be difficult in testing and product selection. The traditional TPC-DS (transaction processing performance council-decision support) benchmark [1], which only has one retail scene, is not sufficient enough for database product selection of other business type. To deal with this drawback, an analytical database testing platform which can plug-in a variety of business scenes was developed. The telecommunication scene was also provided to explain the design concept, system architecture and scene development of this testing platform. The platform provides scene plug-in function which is easy to use without any programming background. Users can select and edit scenes according to their own business type to meet the requirements of analytical database testing and selection in different industries.

Keywords—database, analytical database, database testing, database product selection, big data, distributed system

I. INTRODUCTION

Analytical databases are databases mainly used in OLAP (Online Analytical Processing) related businesses to support complex analysis of massive amount of data. Analytical database products have various architectures, each of which has its own suitable business scenes. The performance indicators of analytical databases are greatly affected by data model, the way of sharding, data volume and configuration tuning of hardware equipment in different scenes. The testing and selection of analytical database has always been a difficulty in industry. As the domestic analytical database market in China enters a period of vigorous development, the demand for testing and selection of analytical database is further increased. A classic benchmark for analytical databases is TPC-DS, a test benchmark with high recognition [1]. However, the business scenes of TPC-DS are limited to the retail industry, using TPC-DS for other scenes (such as finance, telecommunication, etc.) is not ideal. In order to solve the difficulty of analytical database selection of various scenes and make up for the defects of TPC-DS, we have developed an analytical database testing platform. Compared with the TPC-DS testing tool, the biggest advantage and innovation of this platform is the advanced scene plug-in module. This module enables users without software development background to define, modify, and extend scenes to make it more customized. The platform can also generate data for specific scenes according to the user defined data generation rules. We also developed a telecommunication scene which originated from real business scenario in telecommunication industry. The platform has already been used for the testing of multiple database products in product selection.

II. DATABASE INDUSTRY OVERVIEW

Database is a key module of software storage system and an important foundation of big data industry. Since the first hierarchical model database was developed by General electric company in 1961, database products have undergone more than 60 years of continuous evolution and iteration, forming complex and diverse technical architectures [2]. In recent years, driven by both market demands and industry applications, enterprise business system has to face the pressure of large data volume and high concurrency, which puts forward more demands and challenges to database products. On the other hand, the development of computer hardware and big data open source projects provides a new growth point for the technical evolution of database products. Under these circumstances, the domestic database industry has entered a stage of vigorously development. Traditional database vendors accelerate product iteration and constantly introduce new database products. At the same time, database manufacturers such as PingCap and Giant Sequoia continue to emerge. With the popularity of the cloud native concept, many public cloud service providers have also developed their own database products. It is certain that more and more Chinese domestic database products will come into the market. It is estimated that by 2024, the market size of relational database in China will exceed 27 billion RMB, with an average annual growth rate of 20% to 30% per year [3]. The emerging of database products indicates the flourishing of the industry, but it also makes the selection of database products difficult for a variety of users.

III. ANALYTICAL DATABASE TESTING AND SELECTION

Compared with the relatively mature transactional database, the analytical database is difficult in database product selection. Analytical database is mainly used in OLAP, which requires storing massive amount of data and supporting online statistics report, online data analysis, random query and other works to explore the value of information [4-5]. The amount of data in the application scene of OLAP is usually TB or PB level, which is much higher than transactional database application scene. Due to the limited storage and computation capacity of traditional centralized system, analytical databases have to support distributed system. Distributed systems are subject to the CAP (Consistency, Availability, Partition) theorem, which means consistency, availability and partition tolerance cannot be satisfied at the same time, there is no one perfect architecture ideal for all scenarios [6]. Analytic databases, which built on distributed systems, also have different architectures with varies of performance focuses, such as

MPP databases that focus on consistency and Hadoop databases that focus on partition tolerance [7-8]. Analytical databases generally support column storage, where each column of data is organized together in a contiguous physical area on disk. Compared with row storage, column storage has higher efficiency in batch query, aggregation and projection, while lower efficiency in selection, insertion and modification. Some analytical database products, such as Greenplum, support both storage strategy and provide conversions between each other. User can select the appropriate data storage form for their own business scene. In addition, analytical databases also derive various special data models to further optimize the performance of different business scenes: For example, the big table structure [9-10] proposed by Google put all associated fields in one table, which greatly reduces the overhead caused by multi-table associated query. This data model has been applied in HBase and become an important component of the big data ecosystem. The pre-aggregation model represented by Druid optimizes the group-by query operation. In the process of data storage, all dimensions are aggregated, and indexes are created for each dimension, which makes the aggregated query of each dimension much faster. The multidimensional database products represented by Apache Kylin store data in the form of multidimensional arrays, which is also called CUBE structure. The attribute value for all of the dimensions can be used as index for query, which greatly enhances the query efficiency. These data models with specific optimization can greatly improve database performance in

specific business scenes.

The selection of analytical database is highly dependent on business scenes. Different architectures and design concepts make database products have their pros and cons [11]. The real business scenes of today's enterprises are extremely complex and difficult to analyze. The most reliable way to judge whether a database product is suitable or not is to query the database in a scene as real as possible and record the quantitative indicators, such as QPS (query per second). It is important to ensure that the data volume, table structure, test query statements, and concurrency of the test scene are as close to the real business as possible. In terms of analytical database testing and selection, TPC-DS is the test benchmark with high recognition. TPC-DS adopts multi-dimensional data models such as snowflake model and star model. The SQL used for testing includes OLAP operations such as data mining, report generation and online query [1]. TPC-DS played an important role in the development of big data industry. However, with the deepening of big data technology in all business types, scene-specific analytical databases emerge rapidly. It is inappropriate to use the retail scene of TPC-DS to provide a reliable reference for real business scenes, which are usually more complicate and diverse.

IV. PLATFORM SOFTWARE

Considering the difficulties of analytical database testing and selection, we completed the research and development of the analytical database testing platform in 2020. So far, 11

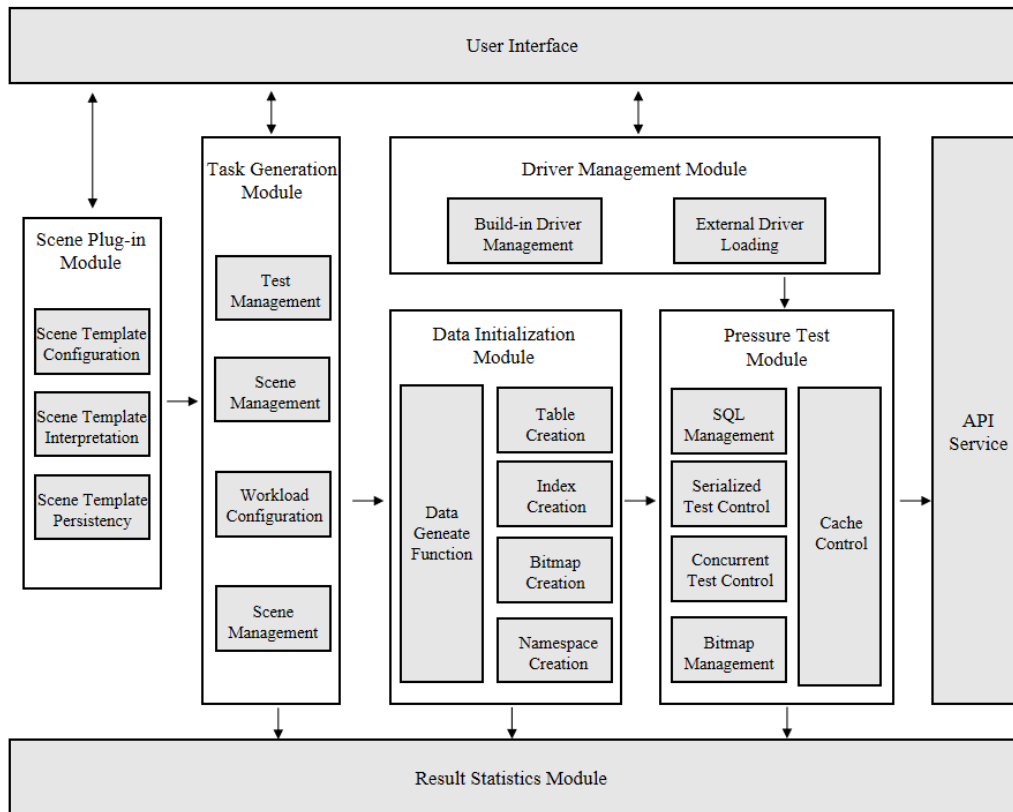


Fig. 1: The overall platform architecture, gray boxes represent micro services

analytical database products have been tested by using this platform.

A. Design concept

The functionality of the platform mainly include two parts. Firstly, the platform needs to generate data for testing according to the configuration of specific business scene. Considering that the data volume in scenes of distributed analytical database is usually at the level of TB or PB [12], the storage capacity of a single node becomes the bottleneck. The testing platform needs to support distributed data generation. Secondly, the platform shall provide query load, support serial and concurrent execution of SQL statements, support monitor and record the usage of basic resources (such as CPU, memory, IO) of the cluster, and evaluate the execution of test statements (such as completion degree, time consuming). The platform is designed to be flexible and easy to be configured to connect to different database product. The platform provides the plug-in function for testing scenes and JDBC(Java Data Base Connectivity) database drivers. Testing scenes can be easily modified and expanded by user to meet their own requirement. The platform has flexible data generation function, which can support data generation of different data types and various statistical distribution types based on user defined data generation rules. The platform contains a simple and user-friendly graphical user interface. User can select the JDBC driver, testing scene and configuration parameters according to the prompts, and conduct guided experiment designing. The development of this platform is based on Java language and Spring framework, with good cross-platform deployment capability. The platform adopts the design scheme of modularization, and made up of multiple micro-services, which lays a foundation for the future container transformation.

B. Platform Architecture

The platform consists of 6 main function modules and 22 micro-services in total(Fig. 1).

a) Driver management module: The platform uses JDBC to access database to be tested. JDBC is an international database application programming interface based on Java, which can establish the connection between Java program and database, send SQL instructions to database and collect the results returned [13-14]. All regular database manufacturers have the implementation of JDBC interface which delivered in the form of JDBC driver package. The module provides driver management function and supports the upload, load, search, change, delete and other operations. Users can flexibly load different drivers according to their test plan without modifying the code, which greatly improves the test efficiency. So far, various databases has been tested including greenplum, mysql, xcloud, tidb, PostgreSQL, etc.

b) Task management module: Task management module mainly provides the function of task configuration, initialization, query, and process monitoring, it also provides the recording functions of test process and raw data. This module support multi-task configuration, each task process is independent from each other. The back-end of this module has MySQL database to record the progress and results of the test in real time to ensure that the data will not be lost due to power failure or other reasons. To better monitoring the test process, prevent fraud, and facilitate user to analyze and

solve the problems encountered, the task management module also provides log generation function. The generated logs include status logs, test progress logs and error logs. The log files are encrypted and stored in the form of TXT document. In order to facilitate users to analyze the log, the module provides log query interface, supports status log query and test progress log query.

c) Data initialization module: The data initialization module has its main function in initializing test parameters and generating data for testing. The data initialization module supports scene selection, data volume configuration, data automatic generation and metadata modification. User can generate test data of different data volume, and adjust the tabular construction statements. The initialization module supports the centralized and distributed generation of data, it can be configured according to the size of the test data and the storage capacity of each individual nodes in cluster. The data initialization module retains a high degree of flexibility in the data import function because different database product varies a lot correspond to data import statements. Users can customize the data import script and upload it to the module.

d) Pressure test module: The main function of the pressure test module is to provide the test load and collect the feedback data from the database under testing. Considering the variety of query statements of each database product, the test module supports statement syntax modification and provides a function for difference comparison. Before generating the load, the module will verify the validity of the testing data and query statements. It will feedback errors if existed, and ensure the normal conduct of the subsequent testing. Pressure test module can provide both serial and concurrent SQL query loads, user can set the number of concurrency for concurrent test. In terms of data collection, the module supports collection of real-time indicators at server level, including execution time of each SQL statement, degree of completion, cluster size, CPU core numbers, CPU utilization, memory size, memory utilization, IO utilization, etc.

e) Result statistics module: The main functions of this module are generating statistics of test results, calculating the indicators and displaying the test results. The platform can output report for data loading time, serial and concurrent test time statistics and SQL completion statistics. The platform can display system status and automatically plot CPU, memory, and IO load curves during the test. In order to facilitate the horizontal comparison of the overall ability of each database, we put forward two comprehensive indicators, namely performance indicator and cost performance indicator. Performance indicators reflect the total workload completed per unit time of the platform, and comprehensively reflected the data scale, concurrent number, SQL completion degree, data loading time, serial test time and concurrent test time in the testing process. The empirical formula is as follows:

$$I = \frac{ANP}{S^3 \sqrt[3]{T_{Load} T_{Serial} T_{Parallel}}} \quad (1)$$

Where I is the performance indicator, A is the configured data volume for testing (in GB), S is the standard data volume for selected scene(in GB), N is the concurrency

degree of concurrent testing, P is the test completion degree of SQL, T_{Load} is the data loading time (in seconds), T_{Serial} is the serial test time (in seconds), $T_{Parallel}$ is the concurrent test time (in seconds). The cost performance indicator reflects the workload completed per unit of time with unit amount of hardware resources. The empirical formula is as follows:

$$C = \frac{I}{S_{CPU} S_{Memory}} \quad (2)$$

Where C is the cost performance indicator, I is the performance indicator, S_{CPU} is the amount of CPU used (the total number of physical threads in cluster multiplied by the percentage of usage), and S_{Memory} is the amount of Memory used (in GB). The calculation of these comprehensive indicators is completed automatically by the platform, which can provide a reference for the horizontal comparison of different database products. To evaluate these indicators and

TABLE I. VALUE OF PERFORMANCE INDICATOR(I) AND COST PERFORMANCE INDICATOR(C) WHEN TESTING POSTGRESQL

Number of Nodes	5	10	20
Performance Indicator (s^{-1})	89714	129487	172103
Cost Performance Indicator($s^{-1}GB^{-1}$)	8.08	4.25	1.78

give a general benchmark, testing has been done using PostgreSQL version 9.2 with data volume 5.5TB. Three independent groups have been tested with cluster size to be 5, 10 and 20 separately (TABLE I). Each node in the cluster is a Huawei Server H22H-05 with 16 cores Intel Xeon Gold 5218 CPU, 8*32G RAM and 6*1.8TB hard drives.

f) *Scene plug-in module*: In order to support various scenes from actual business activities, the platform itself does not contain the specific implementation of testing scenes. Instead, the platform ensures that different test scenes can be flexibly plug-in through the scene plug-in module. The test scenes are defined by CSV file and can be uploaded to the platform by user. The plug-in module of the scene plays the role of interpreter and supports a complete set of rules to define the scene content. The personnel involved in the scene design can complete the configuration of the scene table structure, field information, SQL query statements and

data generation rules through the Excel table without having programming background.

V. TELECOMMUNICATION SCENE DESIGN

The design of analytical database test scenes is a complex but important process. The key of designing is to ensure that the test scenes can represent the main feature of real scenario. For the aspects of table structure, data distribution, data volume and query statement, the test scene needs to be as close as possible to the actual business scene, which should also be typical in the corresponding industry. If the test scene varies a lot from the actual situation. As a data-intensive industry, telecommunications has massive amount of users, who generate massive amount of behavior data in the communication pipeline each day [15-16]. Analytical database products are widely used in analyzing user behaviors in order to gain comprehensive understanding of consumer information. With finely classified customer groups, a more customized actions can then be developed and applied for each group that would meet their specific needs. This user labor processing scene is important and take most of the effort in telecommunication analytic workload. Therefore, we choose the user label processing scene in telecommunication business as the test scene. This test scene integrates the data of mobile users from life to work, from communication to the internet usage, and covers the location and behavior information for 24 hours. This test scene has the characteristics of large data volume, multi-dimensional and wide coverage, which can fully reflect the characteristics of the real situation in the telecommunication industry.

A. Table design

After data masking of tables from actual production environment in telecommunication business, 20 of them were selected as test tables (Fig. 2). The minimum number of fields in the table is 10, the maximum is 130, and the average number of fields is above 80, covering the data of six telecommunication domains, namely, customer domain, product domain, business use domain, accounting domain, location domain and internet domain.

The number of rows of the largest table in the test scene is about 300 million, and the total amount of data generated by the associated sub-table is 20TB, which has the characteristics of large data volume, big table and multi-

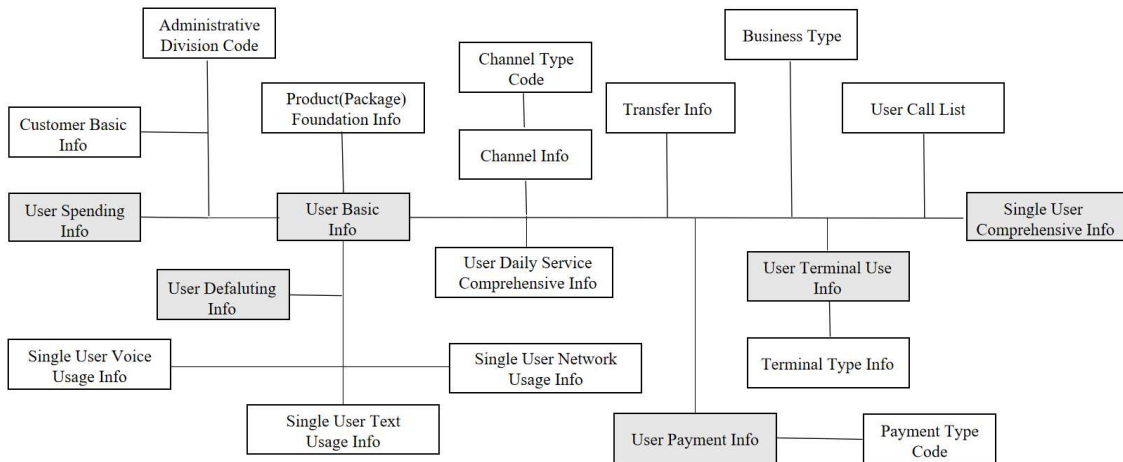


Fig. 2: Overview of telecommunication scene table structure, the gray boxes represent fact tables, the white boxes represent dimension tables

dimensional. The test tables contain the fact tables and dimension tables required for the telecommunication analytical scene. A fact table stores fact data that is used to describe specific events within a business. The dimension table is the feature of the fact in the fact table, which can be used as the window for data analysis. The relation between fact record and dimension attribute is represented by the combination of star model and snowflake model. In the ER (entity-relationship) model of the test scene, there are 6 fact tables, namely the user basic information table, the single user comprehensive information table, the user payment information table, the user spending information table, the user defaulting information table and the user terminal use information table.

B. Test statement design

The key points of the test statement designing is to ensure that the set of the statement can cover the data processing workflow of the entire telecommunication scene(Fig. 3), which mainly includes the following four aspects:

- The data processing department completes the creation of business tables and data initialization.
- According to the scene model and initial data, the data processing department designs query SQL, binds analysis logic, aggregates business rules and strategies, and finally processes and generates results.
- The data processing department synchronizes the generated recommendation results to a distributed data storage system that supports high concurrency queries.
- The business department initiates the query demand, passes the query condition to the data service department, and obtains the final query result from the data service department.

After query statements screening and combining query patterns with similar structure, 20 SQL query statements were extracted from common query statements in data processing workflow as test statements(TABLE II). These SQL statements include statistical query statements, report queries, exploratory research query, specific condition

queries, etc., covering the operation, business and management aspects of telecommunication services.

TABLE II. SQL STATEMENTS FOR TELECOMMUNICATION SCENE TESTING

Operation
SQL1-User call behavior statistics
SQL2-Toll type user statistics
SQL3-Product(package) usage query
SQL10-Product statistics with most/least users
SQL11-Profit statistics from each province
SQL12-Profit statistics from a province in certain time range
SQL14-proportion of overdue users
SQL16-statistics of most valuable product
SQL17-statistics of most valuable channel
Business
SQL13-Payment from each user group
SQL15-User who pay and renew program
SQL18-Market share for each terminal
SQL19-The cellphone brand which has the largest number of user
SQL20-Device replacement frequency for each user group
Management
SQL4-User feature information query
SQL5-User number related information
SQL6-Get all numbers from one user
SQL7-City with largest user increasing rate
SQL8-Channel information with largest number of new users
SQL9-User transferred from other operators

The complexity of data processing workflow makes it inevitable for these test SQL statements to contain some complex query operations such as multi-dimensional roll-up or drill-down and multi-table aggregation. These operations need to be completed within the time limit during the test, which raises the threshold and also provides an important indicator in product screening.

VI. OUTLOOK

With the rapid development of big data technology, the importance of data has been widely recognized, and data analysis has become an indispensable driving force for the development of any organization [17]. The user groups of the analytical database is not only restricted to large Internet enterprises and high-tech companies, but also traditional industries such as telecommunications, medical care, finance, education, transportation, catering, geographic information and urban management [18-24]. The database testing platform which can be connected to a variety of scenes provides great convenience for the database testing and product selection from diversified user groups. As the future work, we will continue to improve the functions of the testing platform, strengthen its ability in multi-scene access, and improve the data generation rules to support more abundant data distribution patterns. We will also complete the container transformation of the test platform, make it more convenient for users to deploy and use, especially for cloud deployment. On the other hand, we will cooperate with leading enterprises from various industry types to develop more abundant test scenes.

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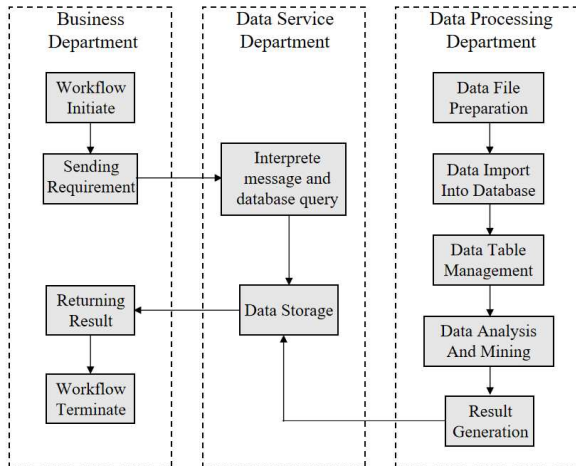


Fig. 3: Data processing workflow in telecommunication industry

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